

# Deepak Sathyan

## Curriculum Vitae

[arXiv](#) | [InspireHEP](#) | [Google Scholar](#) | [ORCID](#) | [Website](#)

✉ dsathyan@tamu.edu

📍 Mitchell Institute for Fundamental Physics and Astronomy, Texas A&M University, College Station, TX

## Education

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### University of Maryland

College Park, MD

#### Ph.D. in Physics

Aug 2018 – July 2024

Thesis: Unifying Searches for New Physics with Precision Measurements of the W

Boson Mass

Advisor: Dr. Kaustubh Agashe

### Boston University

Boston, MA

#### Bachelor of Arts in Physics, Summa Cum Laude with Honors

Aug 2014 – May 2018

GPA: 3.9

Minor: Mathematics

Thesis: Muon (g-2): Searching for the Muon Electric Dipole Moment

Advisor: Dr. Robert Carey

## Experience

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### Postdoctoral Research Associate

College Station, TX

Mitchell Institute for Fundamental Physics and Astronomy

Sept 2024 – Aug 2027

Texas A&M University

3 years

## Research Interests

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### Beyond the Standard Model searches with Collider Physics

searches at the LHC, sub-GeV dark matter, invisible new physics via precision measurements, Kaluza-Klein modes in Randall-Sundrum models, baryon number violation models, muon collider phenomenology

### Searches for sub-GeV Dark Matter

Constraints on dark matter-neutrino interactions, dark photon production at the LHC, dark sector production at muon collider, astrophysical signals of cosmic ray-dark matter scattering, astrophysical sources of sub-GeV dark matter

### Neutrino Physics

Supernova neutrino attenuation and detection, neutrino oscillations in open systems, neutrinophilic mediators

## Publications

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### 5. New laboratory constraints on neutrinophilic mediators

July 2024

P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang

[arxiv.org/abs/2407.12738](https://arxiv.org/abs/2407.12738) (Phys.Lett.B 868 (2025) 139765)

- 4. ‘Unification’ of BSM searches and SM measurements: the case of lepton+MET and  $m_W$**

K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan  
[arxiv.org/abs/2404.17574](https://arxiv.org/abs/2404.17574) (JHEP 02 (2025) 139)

Apr 2024
- 3. A new purpose for the W-boson mass measurement: Searching for New Physics in lepton+MET**

K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan  
[arxiv.org/abs/2310.13687](https://arxiv.org/abs/2310.13687) (Phys.Lett.B 855 (2024) 138774)

Oct 2023
- 2. Energy-peak based method to measure top quark mass via B-hadron decay lengths**

K. Agashe, S. Airen, R. Franceschini, J. Incandela, D. Kim, D. Sathyan  
[arxiv.org/abs/2212.03929](https://arxiv.org/abs/2212.03929) (JHEP 06 (2023) 021)

Dec 2022
- 1. LHC Signals for KK Graviton from an Extended Warped Extra Dimension**

K. Agashe, M. Ekhaterachian, D. Kim, D. Sathyan  
[arxiv.org/abs/2008.06480](https://arxiv.org/abs/2008.06480) (JHEP 11 (2020) 109)

Aug 2020

## Preprints

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- 4. Intensity-Frontier Signals of Warped Extra Dimensions**

D. Kim, D. Sathyan, A. Verma  
[arxiv.org/abs/2608.20482](https://arxiv.org/abs/2608.20482)

Aug 2026
- 3. Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering**

B. Dutta, D. Goswami, J. Kumar, M. Rai, D. Sathyan  
[arxiv.org/abs/2605.08010](https://arxiv.org/abs/2605.08010)

May 2026
- 2. A Baryon and Lepton Number Violation Model Testable at the LHC**

A. Bhoonah, F. Burk, D. Liu, T. Ou, D. Sathyan  
[arxiv.org/abs/2508.21064](https://arxiv.org/abs/2508.21064)

Aug 2025
- 1. New Constraints on Neutrino-Dark Matter Interactions: A Comprehensive Analysis**

P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang  
[arxiv.org/abs/2507.01000](https://arxiv.org/abs/2507.01000)

July 2025

## Articles in Preparation

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- 1. LHC as a Beam Dump: Probing Light Dark Photons**

B. Dutta, A. Karthikeyan, D. Kim, H. Kim, T. Kim, D. Sathyan  
[arxiv.org/a/sathyan\\_d\\_1](https://arxiv.org/a/sathyan_d_1)

Sept 2026
- 2. New Probes of Dark Sectors at a Muon Collider**

B. Dutta, A. Karthikeyan, D. Kim, K. Kelly, D. Sathyan  
[arxiv.org/a/sathyan\\_d\\_1](https://arxiv.org/a/sathyan_d_1)

Jan 2027

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| <p><b>3. Exploring open systems approach to neutrino oscillations</b><br/> A. Chandra Shekar, K. Kelly, D. Sathyan, L. Strigari, T. Zhou<br/> <a href="https://arxiv.org/a/sathyan_d_1">arxiv.org/a/sathyan_d_1</a></p>                                        | <p>Mar 2027</p> |
| <p><b>4. LHC Signals for KK Gravitons Producing 8-Particle Final States</b><br/> K. Agashe, D. Kim, P. Maksimovic, S. Mondal, M. Osherson, K. Panchal, D. Plotnikov, D. Sathyan<br/> <a href="https://arxiv.org/a/sathyan_d_1">arxiv.org/a/sathyan_d_1</a></p> | <p>Mar 2027</p> |

## Snowmass2021 Contributions ---

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| <p><b>5. TF07 Snowmass Report: Theory of Collider Phenomena</b><br/> F. Maltoni et al.<br/> <a href="https://arxiv.org/abs/2210.02591">arxiv.org/abs/2210.02591</a></p>                                                                                                                           | <p>Oct 2022</p>  |
| <p><b>4. Report of the Topical Group on Top quark physics and heavy flavor production for Snowmass 2021</b><br/> K. Agashe et al.<br/> <a href="https://arxiv.org/abs/2209.11267">arxiv.org/abs/2209.11267</a></p>                                                                                | <p>Sept 2022</p> |
| <p><b>3. Report of the Topical Group on Physics Beyond the Standard Model at Energy Frontier for Snowmass 2021</b><br/> T. Bose et al.<br/> <a href="https://arxiv.org/abs/2209.13128">arxiv.org/abs/2209.13128</a></p>                                                                           | <p>Sept 2022</p> |
| <p><b>2. Snowmass2021 - White Paper, Implications of Energy Peak for Collider Phenomenology: Top Quark Mass Determination and Beyond</b><br/> K. Agashe, S. Airen, R. Franceschini, D. Kim, D. Sathyan<br/> <a href="https://arxiv.org/abs/2204.02928">arxiv.org/abs/2204.02928</a></p>           | <p>Apr 2022</p>  |
| <p><b>1. Snowmass2021 White Paper: Collider Physics Opportunities of Extended Warped Extra-Dimensional Models</b><br/> K. Agashe, J.H. Collins, P. Du, M. Ekhterachian, S. Hong, D. Kim, R.K. Mishra, D. Sathyan<br/> <a href="https://arxiv.org/abs/2203.13305">arxiv.org/abs/2203.13305</a></p> | <p>Mar 2022</p>  |

## Muon g-2 Articles ---

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| <p><b>3. The straw tracking detector for the Fermilab Muon g-2 Experiment</b><br/> B.T. King et al.<br/> <a href="https://arxiv.org/abs/2111.02076">arxiv.org/abs/2111.02076</a> (JINST 17 (2022) P02035)</p>                                                          | <p>Nov 2021</p> |
| <p><b>2. Beam dynamics corrections to the Run-1 measurement of the muon anomalous magnetic moment at Fermilab</b><br/> Muon g-2 Collaboration<br/> <a href="https://arxiv.org/abs/2104.03240">arxiv.org/abs/2104.03240</a> (Phys.Rev.Accel.Beams 24 (2021) 044002)</p> | <p>Apr 2021</p> |
| <p><b>1. Measurement of the Positive Muon Anomalous Magnetic Moment to 0.46 ppm</b><br/> Muon g-2 Collaboration<br/> <a href="https://arxiv.org/abs/2104.03281">arxiv.org/abs/2104.03281</a> (Phys.Rev.Lett. 126 (2021) 141801)</p>                                    | <p>Apr 2021</p> |

## Ph.D. Thesis

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- 1. Unifying Searches for New Physics with Precision Measurements of the W Boson Mass** 2024  
D. Sathyan  
[drum.lib.umd.edu/items/f9b081c8-444a-4af2-b69e-c8156d11a2c9](https://drum.lib.umd.edu/items/f9b081c8-444a-4af2-b69e-c8156d11a2c9) (Ph.D. Thesis)

## Seminars

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- 9. HEP Seminar at Northwestern University** Sept 2026  
Cosmic Ray–Dark Matter scattering explanation of the Galactic Center Excess
- 8. Fermilab Cosmic Physics Center Seminar** Sept 2026  
Cosmic Ray–Dark Matter scattering explanation of the Galactic Center Excess
- 7. CMS Exotica general meeting** May 2024  
Unification of Searches and Measurements: Probing BSM with the W boson mass measurement
- 6. Seminar at Texas A&M** Dec 2023  
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET
- 5. Seminar at UT Austin** Dec 2023  
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET
- 4. CMS SUSY general meeting** Nov 2023  
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET
- 3. HEP Seminar at University of Pittsburgh** Nov 2023  
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET
- 2. HEP Seminar at Northwestern University** Oct 2023  
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET
- 1. Theory Seminar at Washington University in St. Louis** Oct 2023  
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

## Conference Talks

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- 14. Light Dark World 2026 at Carleton University** July 2026  
Can the LHC be sensitive to the Light Dark World?
- 13. CETUP\* Workshop** June 2026  
Producing the GeV Galactic Center Excess via Cosmic Ray–Dark Matter Scattering

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| <b>12. Particle Physics on the Plains at the University of Kansas</b><br>A Baryon and Lepton Number Violation Model Testable at the LHC                     | Nov 2025  |
| <b>11. CETUP* Workshop</b><br>Can the LHC be sensitive to light dark mediators?                                                                             | June 2025 |
| <b>10. Phenomenology Conference at the University of Pittsburgh</b><br>Can the LHC be sensitive to light dark mediators?                                    | May 2025  |
| <b>9. Particle Physics on the Plains at the University of Kansas</b><br>A comprehensive analysis of supernova neutrino-dark matter interactions             | Nov 2024  |
| <b>8. Texas TACOS @ UT Austin</b><br>A comprehensive analysis of supernova neutrino-dark matter interactions                                                | Oct 2024  |
| <b>7. NuFact 2024 Conference at Argonne National Laboratory</b><br>A comprehensive analysis of supernova neutrino-dark matter interactions                  | Sept 2024 |
| <b>6. Mitchell Conference at Texas A&amp;M University</b><br>A comprehensive analysis of supernova neutrino-dark matter interactions                        | May 2024  |
| <b>5. Particle Physics on the Plains at the University of Kansas</b><br>A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET     | Oct 2023  |
| <b>4. Phenomenology Conference at the University of Pittsburgh</b><br>Probing Dark Matter-Neutrino Interactions via Supernova Neutrinos                     | May 2023  |
| <b>3. Phenomenology Conference at the University of Pittsburgh</b><br>Model-Independent Measurement of Top Quark Mass Using B-Hadron Decay Lengths (Part I) | May 2022  |
| <b>2. April APS Meeting</b><br>Signals of KK graviton from extended warped extra dimensions at the LHC (II)                                                 | Apr 2021  |
| <b>1. Phenomenology Conference at the University of Pittsburgh</b><br>Signals of KK graviton from extended warped extra dimensions at the LHC (II)          | May 2020  |

## Conferences Organized

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**The Mitchell Conference on Collider, Dark Matter, and Neutrino Physics 2026**  
Texas A&M University, May 27-30, 2026

**The Mitchell Conference on Collider, Dark Matter, and Neutrino Physics 2025**  
Texas A&M University, May 13-16, 2025

## Awards

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| <b>Breakthrough Prize in Fundamental Physics</b><br>Muon g-2 Collaboration | Apr 2026 |
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## Schools Attended

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**GGI Lectures on the Theory of Fundamental Interactions**

Jan 2023 – Jan 2023

**TASI 2024: The Frontiers of Particle Theory**

June 2024 – June 2024

## Outreach

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**BU Student Mentorship Program PRISM (2016-2018)**

Mentored incoming freshman undergraduates in physics, providing guidance on courses, research opportunities, and resources

**TAMU Physics and Engineering Festival (2025-2027)**

Presented demonstrations of fluid dynamics to the general public, including high school students and families

**University of South Dakota talk: "My path through particle physics research so far" (December 2024)**

Presented my path through particle physics research to students, showing a variety of interesting ideas to pursue

## Skills

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**Programming:**

Mathematica, Python, C++

**Physics Software:**

MadGraph5\_aMC@NLO, Pythia6, Pythia8, Delphes, ROOT, GEANT4, FeynRules